Superconductivity, Nambu representation and Majorana physics

Today’s learning outcomes

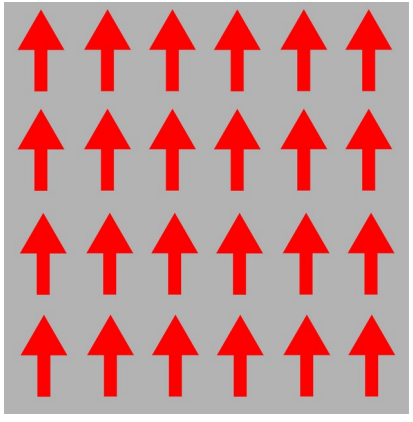
- Recognize the nature of the electronic structure of a superconductor
- Use the Nambu representation to describe superconducting excitations
- Rationalize the emergence of Majorana excitations in topological superconductors

A reminder from previous sessions

Electronic interactions are responsible for symmetry breaking

Broken
time-reversal symmetry

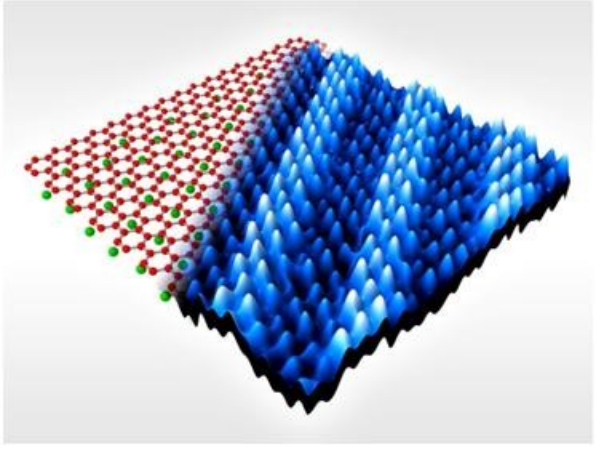
Classical magnets



$$\mathbf{M} \rightarrow -\mathbf{M}$$

Broken
crystal symmetry

Charge density wave



$$\mathbf{r} \rightarrow \mathbf{r} + \mathbf{R}$$

Broken
gauge symmetry

Superconductors



$$\langle c_\uparrow c_\downarrow \rangle \rightarrow e^{i\phi} \langle c_\uparrow c_\downarrow \rangle$$

Quantum matter with interactions

We can consider two broad groups of interacting quantum matter

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$$

With a mean field description

$$H \approx \sum_{ij} \bar{t}_{ij} c_i^\dagger c_j + \sum_{ij} \Delta_{ij} c_i c_j$$

Approximate quadratic Hamiltonian
Effective single particle description

Weakly correlated matter

Without a mean field description

????

No good quadratic approximation
Requires exact solutions or numerical

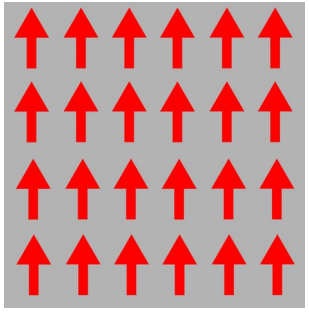
Strongly correlated matter

Correlations and mean field

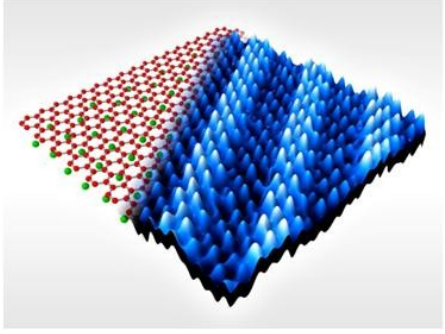
Many quantum states can be approximately described by mean field theories

$$H \approx \sum_{ij} \bar{t}_{ij} c_i^\dagger c_j + \sum_{ij} \Delta_{ij} c_i c_j + h.c.$$

Magnets



Charge density waves



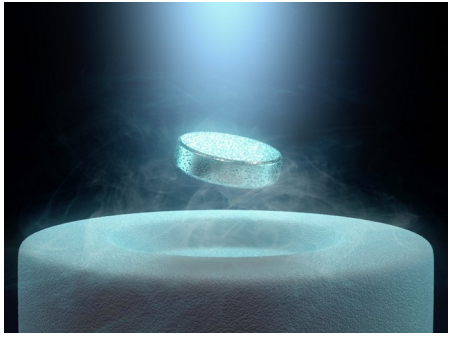
Superconductors



Forms of orrelated matter

Superconductors

with a mean field description

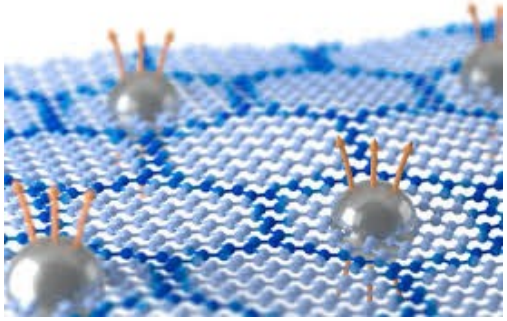


Attractive interactions
Destroyed by magnetic fields

Majorana excitations

Fractional quantum Hall states

without a mean field description

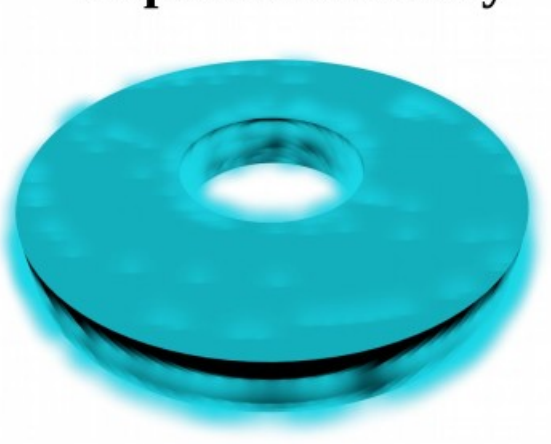


Repulsive interactions
Require strong magnetic fields

Fractional charged excitations

Macroscopic quantum phenomena

Superconductivity

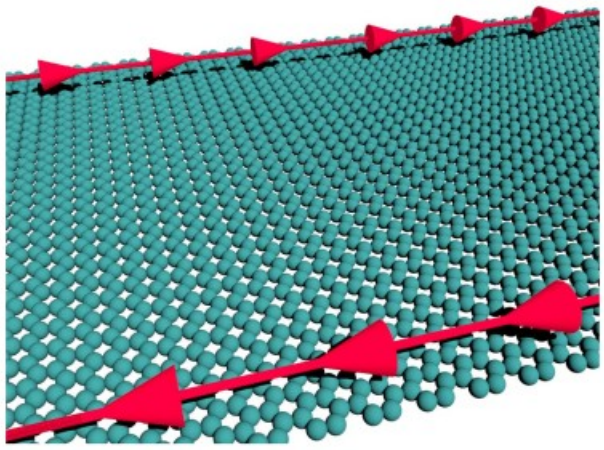


$$\Phi = \frac{h}{2e}$$

Many-body state

Quantization of flux

Quantum Hall effect



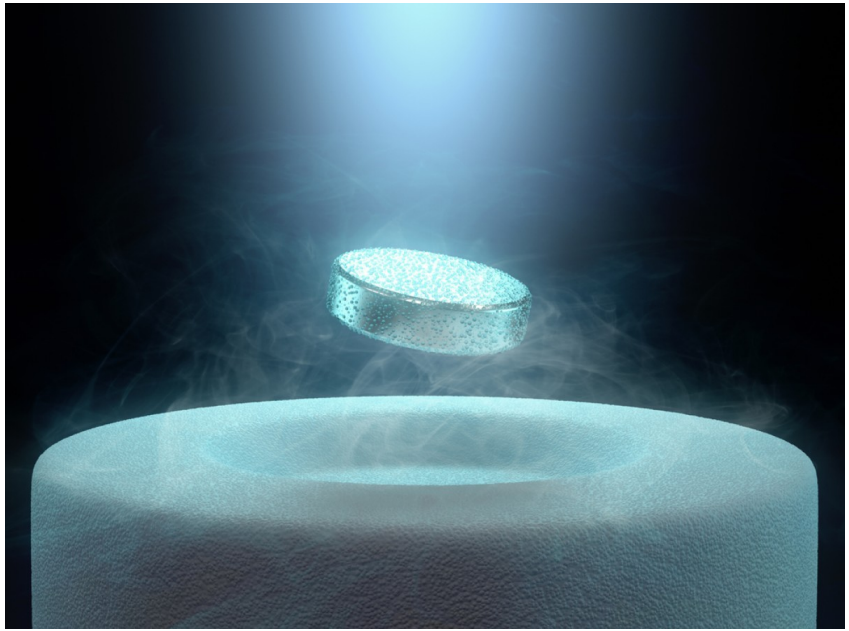
$$\sigma_{xy} = \nu \frac{e^2}{h}$$

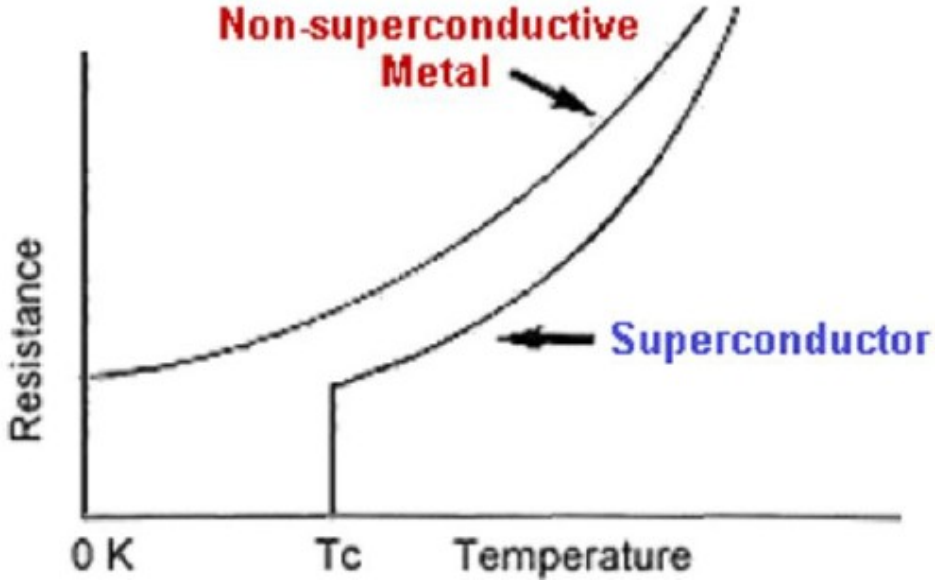
Single-particle state

Quantization of conductance

The theoretical description of superconductivity

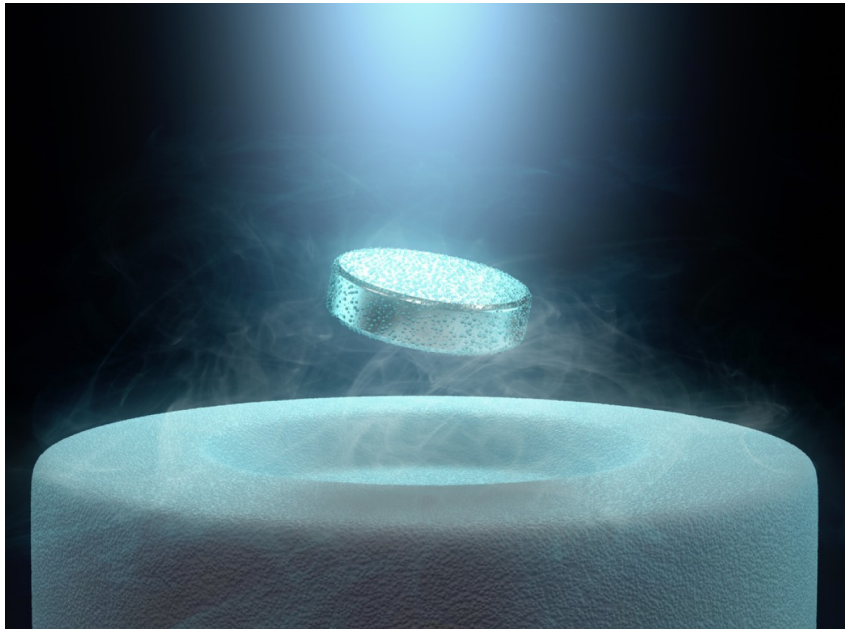
What is a superconductor?

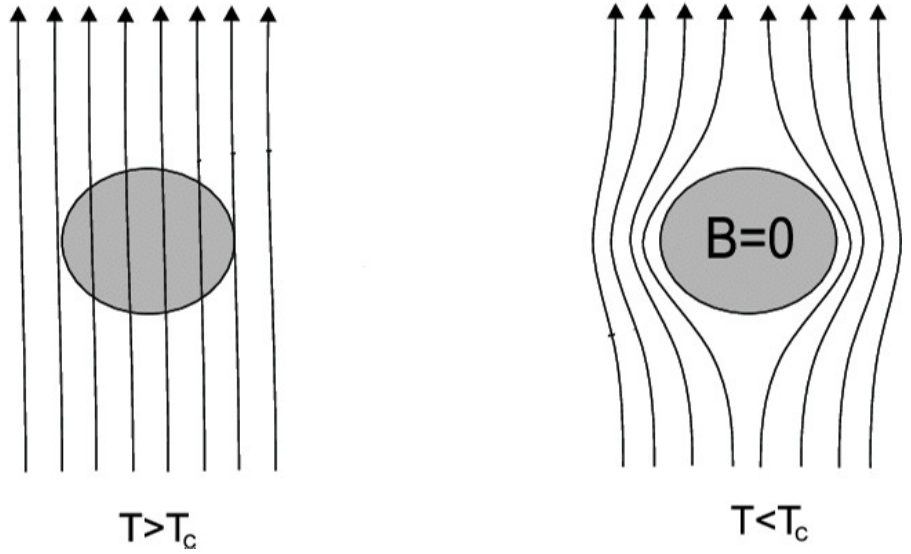




The resistance suddenly drops to 0 at a certain temperature

What is a superconductor?





The magnetic field lines get expelled in the superconducting state

Interactions and mean field

Free Hamiltonian ***Interactions***

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$$

What are these interactions coming from?

- Electrostatic (repulsive) interactions
- Mediated by other quasiparticles (phonons, magnons, plasmons,...)

The net effective interaction can be attractive or repulsive

Superconductivity requires effective attractive interactions

Origin of attractive interactions

Interactions between electrons can be effectively attractive when mediated by other quasiparticles

Conventional superconductors

Phonons

Unconventional superconductors

Antiferromagnetic magnons

Ferromagnetic magnons

Plasmons

Valence fluctuations

Charge fluctuations

...

A simple interacting Hamiltonian

Free Hamiltonian ***Interactions (Hubbard term)***

$$H = \sum_{ij} t_{ij} [c^\dagger_{i\uparrow} c_{j\uparrow} + c^\dagger_{i\downarrow} c_{j\downarrow}] + \sum_i U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$$

From now on lets consider we have a spin degree of freedom $\uparrow, \downarrow$

What is the ground state of this Hamiltonian?

$U < 0$ Superconductivity

$U > 0$ Magnetism

The mean-field approximation, superconductivity

Mean field: *Approximate four fermions by two fermions times expectation values*

Four fermions (not exactly solvable) ***Two fermions (exactly solvable)***

$$U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} \approx U \langle c^\dagger_{i\uparrow} c^\dagger_{i\downarrow} \rangle c_{i\uparrow} c_{i\downarrow} + h.c.$$

$$U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} \approx \Delta c_{i\uparrow} c_{i\downarrow} + h.c.$$

For $U < 0$

i.e. attractive interactions

$\Delta \sim \langle c^\dagger_{i\uparrow} c^\dagger_{i\downarrow} \rangle$ is the superconducting order

A Hamiltonian for a superconductor

Free Hamiltonian ***Pairing term***

$$H = \sum_{ij} t_{ij} [c^{\dagger}_{i\uparrow} c_{j\uparrow} + c^{\dagger}_{i\downarrow} c_{j\downarrow}] + \Delta \sum_{i} c_{i\uparrow} c_{i\downarrow} + h.c.$$

Lets have a look to the term $\Delta c_{i\uparrow} c_{i\downarrow}$

Superconductivity and symmetries

$$H \sim \Delta c_{i\uparrow} c_{i\downarrow} + h.c.$$

$$H|GS\rangle = E_{GS}|GS\rangle$$

This term destroys two electrons

$|GS\rangle$ Ground state

The ground state can not have a well defined number of electrons

$$|GS\rangle \sim |2e\rangle + |4e\rangle + |6e\rangle + ...$$
$$|GS\rangle \sim |1e\rangle + |3e\rangle + |5e\rangle + ...$$

Gauge symmetry and superconductivity

What we know from quantum mechanics

"The phase of a wavefunction (field operator) does not have physical meaning"

This is what we know as gauge symmetry

$$c_n \rightarrow e^{i\phi} c_n$$
$$c_n^\dagger \rightarrow e^{-i\phi} c_n^\dagger$$

Terms in the Hamiltonian that do not change under this transformation

$c_n^\dagger c_m$	$c_{n,\uparrow}^\dagger c_{n,\uparrow} - c_{n,\downarrow}^\dagger c_{n,\downarrow}$	$c_{n,\uparrow}^\dagger c_{m,\downarrow}$	$c_{n,\uparrow}^\dagger c_{m,\uparrow} c_{n,\downarrow}^\dagger c_{m,\downarrow}$
Hopping	*Magnetism*	*Spin-orbit coupling*	*Interactions*

Superconductivity and gauge symmetry breaking

Gauge symmetry

$$c_n \to e^{i\phi} c_n \qquad c_n^\dagger \to e^{-i\phi} c_n^\dagger$$

How does the superconducting order transform under a gauge transformation?

$$\Delta = \langle c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger \rangle$$

Superconductivity and gauge symmetry breaking

Gauge symmetry

$$c_n \to e^{i\phi} c_n \qquad c_n^\dagger \to e^{-i\phi} c_n^\dagger$$

How does the superfluid density transform under a gauge transformation?

$$\Delta \to e^{-2i\phi} \Delta$$

A superconductor breaks gauge symmetry

The electronic structure of a superconductor

The Nambu representation

How do we solve a Hamiltonian of the form $H = \sum_{\mathbf{k},s} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k},s} c_{\mathbf{k},s} + \sum_{\mathbf{k}} \Delta c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow} + h.c.$

Define a Nambu spinor $\Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{\mathbf{k}\downarrow} \\ c^{\dagger}_{-\mathbf{k}\downarrow} \\ -c^{\dagger}_{-\mathbf{k}\uparrow} \end{pmatrix}$ ← Electron sector

← Hole sector

The Hamiltonian in the Nambu basis is quadratic and can be diagonalized

$$H = \Psi^{\dagger}_{\mathbf{k}} \mathcal{H} \Psi_{\mathbf{k}}$$

A single orbital superconductor

The original Hamiltonian $$H = \sum_{\mathbf{k},s} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k},s} c_{\mathbf{k},s} + \sum_{\mathbf{k}} \Delta c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow} + h.c.$$

Can be rewritten as

$$H = \frac{1}{2} \Psi^{\dagger}_{\mathbf{k}} \mathcal{H} \Psi_{\mathbf{k}} \qquad \Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{\mathbf{k}\downarrow} \\ c^{\dagger}_{-\mathbf{k}\downarrow} \\ -c^{\dagger}_{-\mathbf{k}\uparrow} \end{pmatrix}$$

← Electron sector

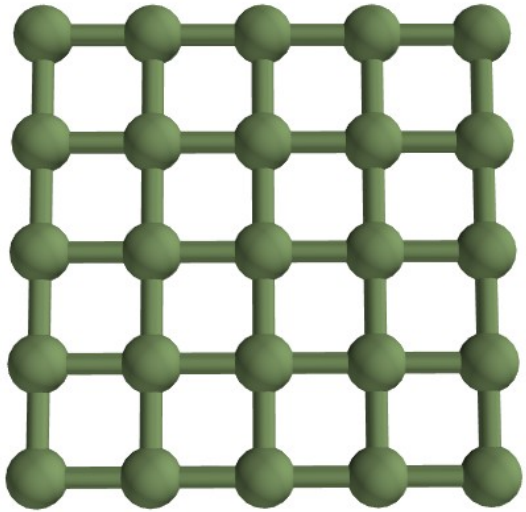
← Hole sector

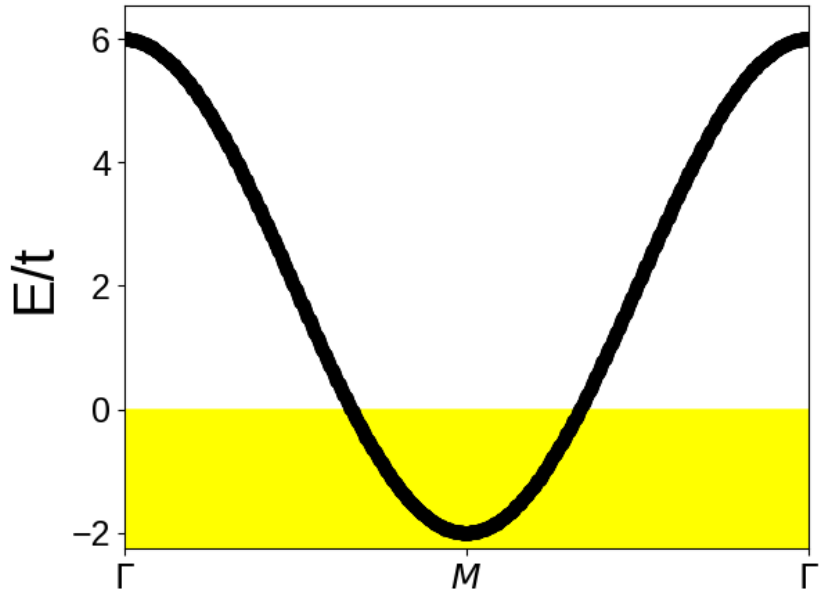
with $$\mathcal{H} = \begin{pmatrix} \epsilon_{\mathbf{k}} & 0 & \Delta & 0 \\ 0 & \epsilon_{\mathbf{k}} & 0 & \Delta \\ \Delta & 0 & -\epsilon_{\mathbf{k}} & 0 \\ 0 & \Delta & 0 & -\epsilon_{\mathbf{k}} \end{pmatrix}$$

A single orbital superconductor

Single orbital in the square lattice

$$H = \sum_{\mathbf{k},s} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k},s} c_{\mathbf{k},s}$$

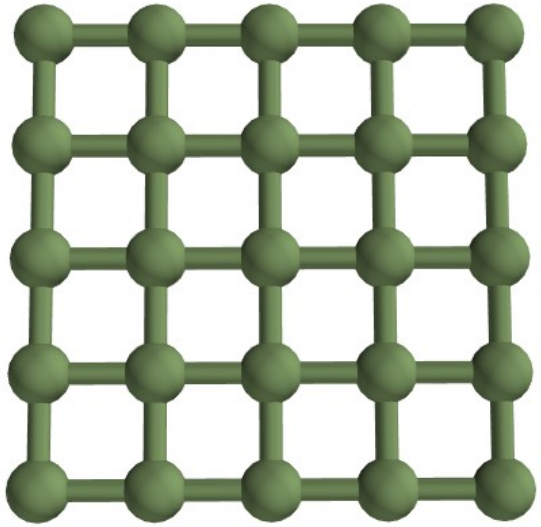




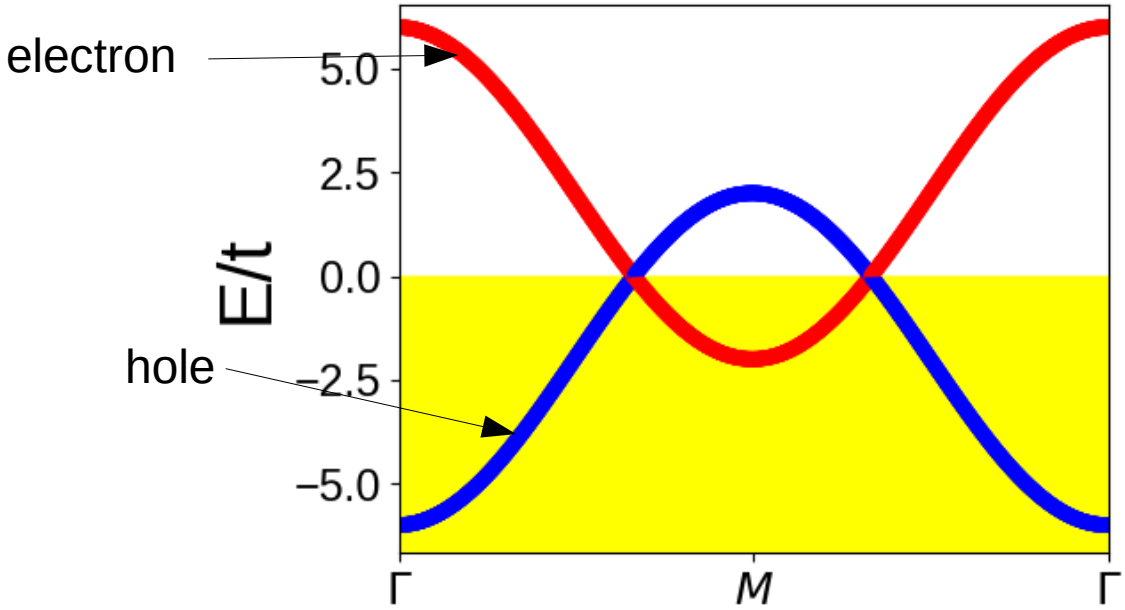
A single orbital superconductor

Single orbital in the square lattice

$$H = \sum_{\mathbf{k},s} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k},s} c_{\mathbf{k},s} + \sum_{\mathbf{k}} \Delta c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow} + h.c.$$



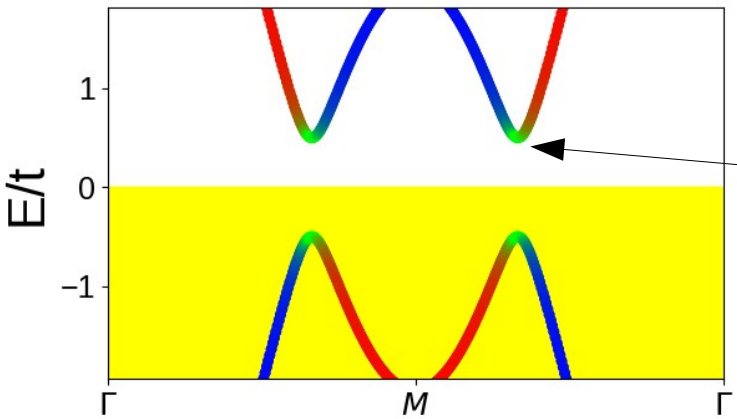
$\Delta \to 0$



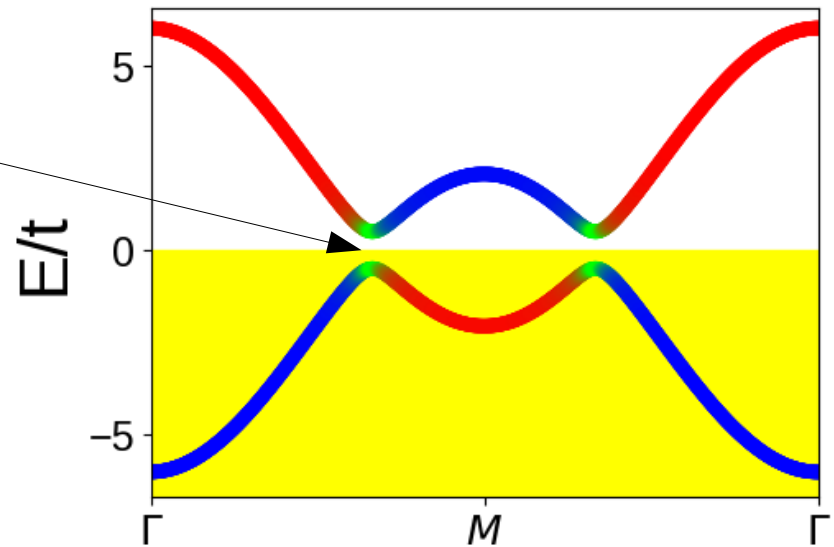
A single orbital superconductor

Single orbital in the square lattice

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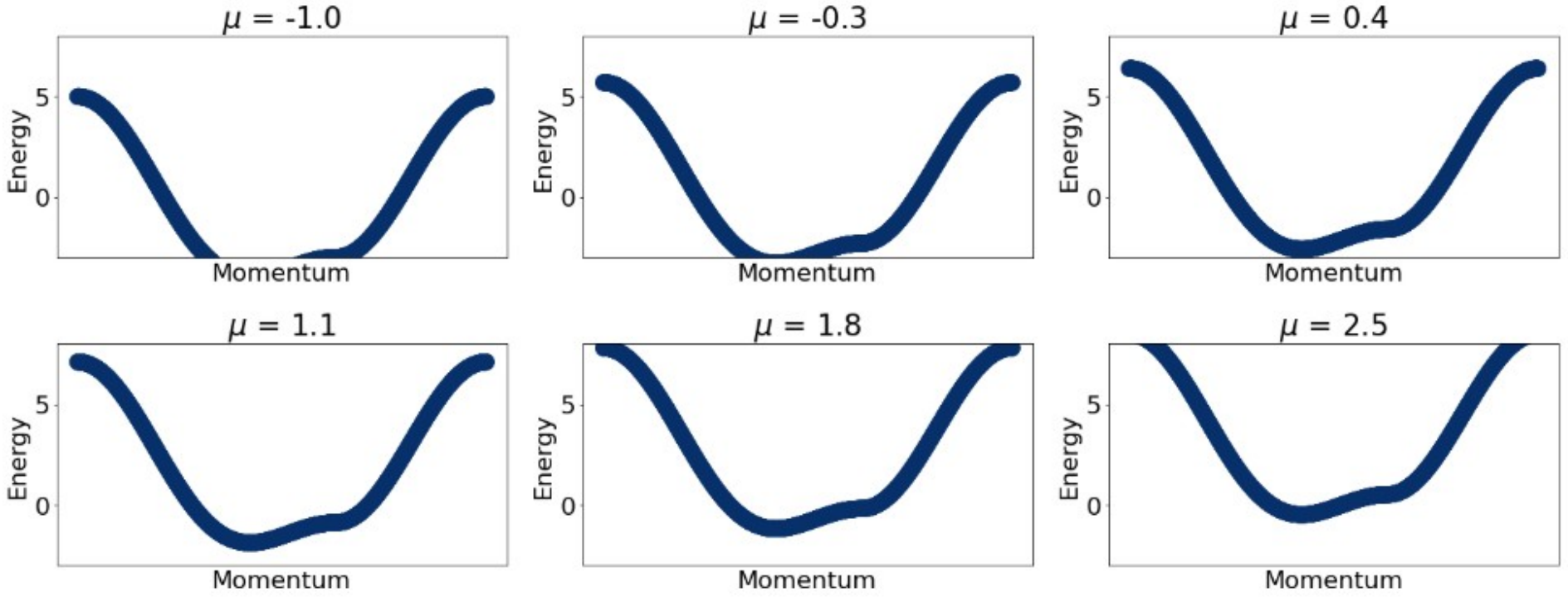


SC gap



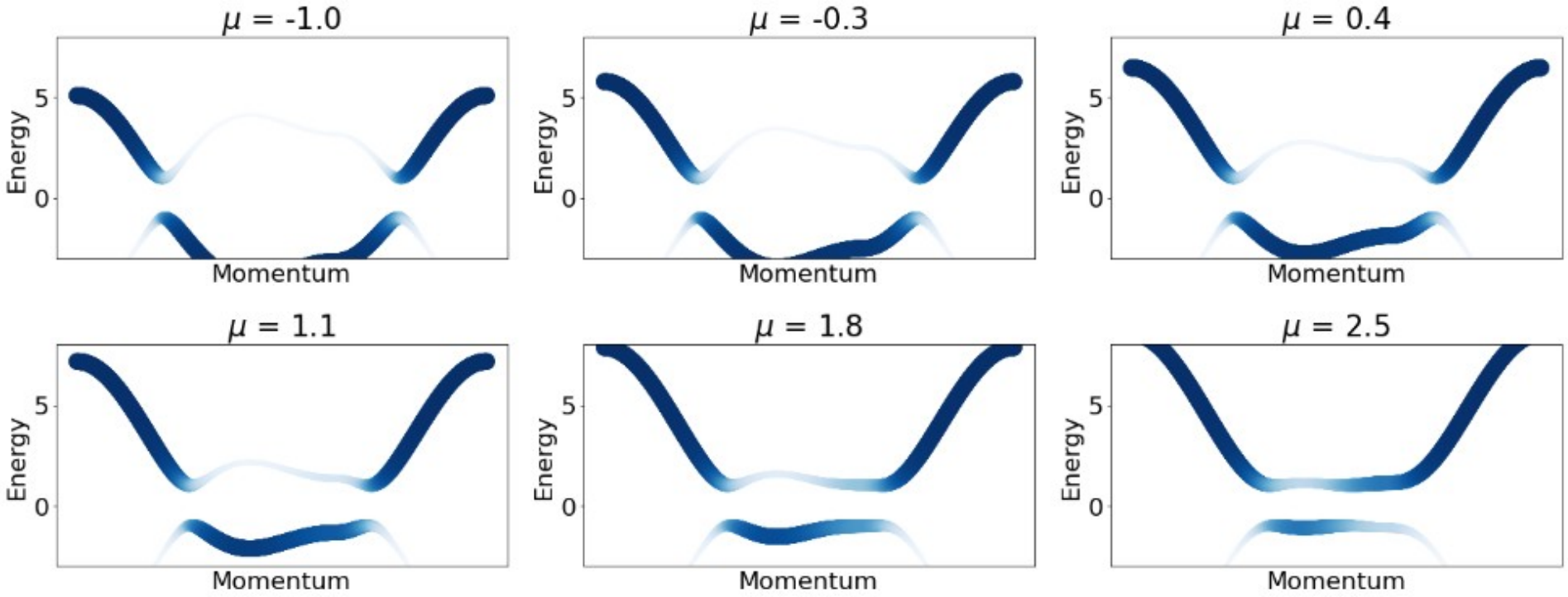
Impact of superconductivity in the electronic structure

Let us take the electronic structure of a triangular lattice



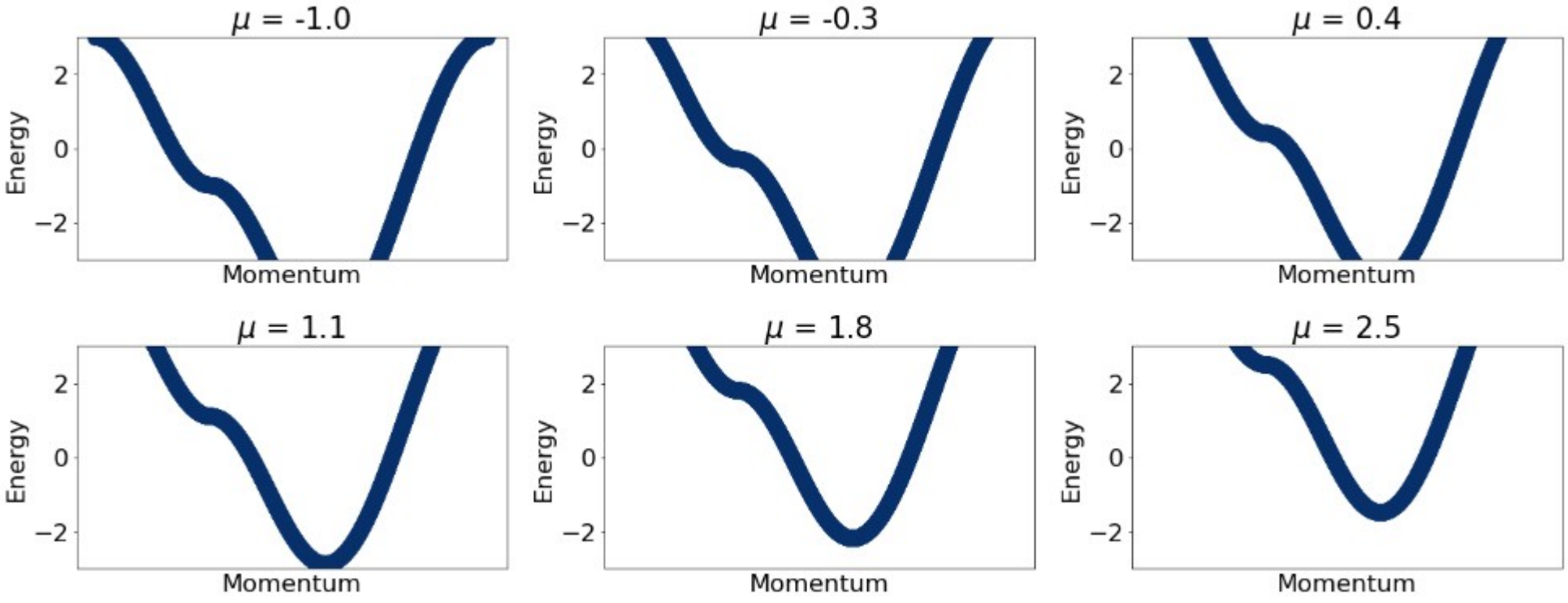
Impact of superconductivity in the electronic structure

A uniform superconducting terms opens up a gap in the electronic structure



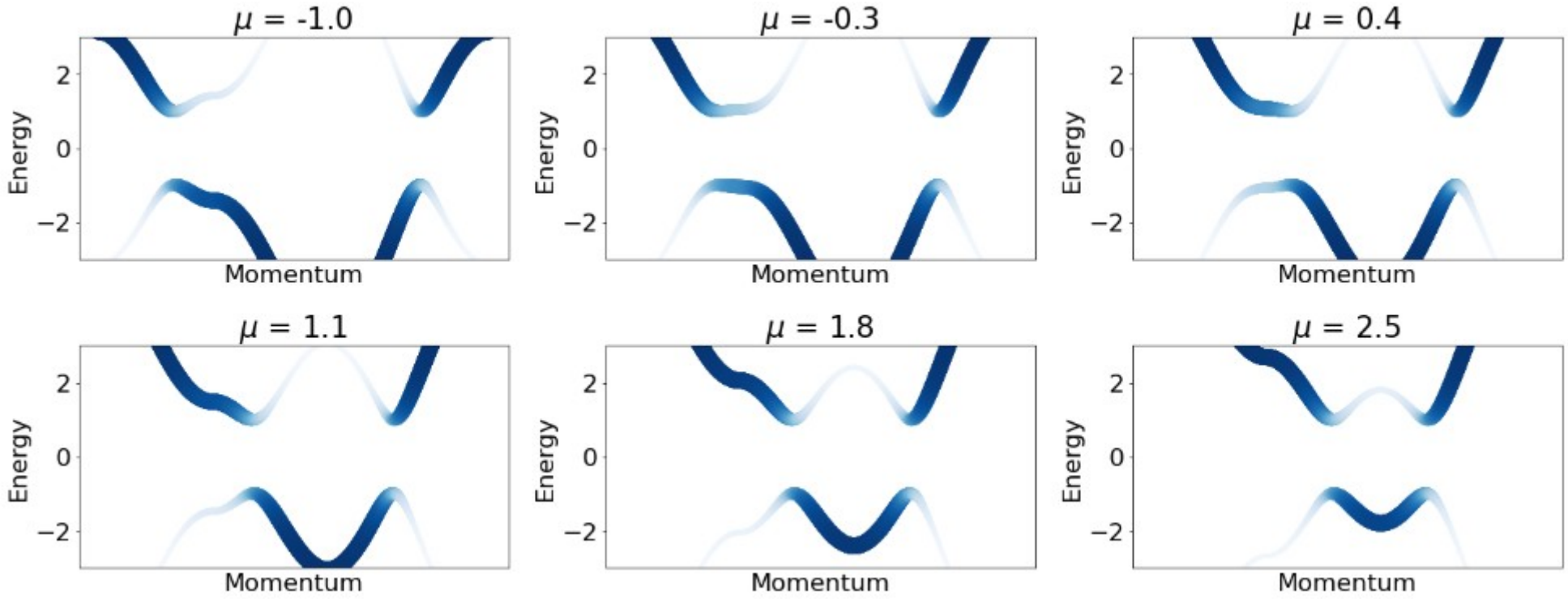
Impact of superconductivity in the electronic structure

Let us take the electronic structure of a square lattice



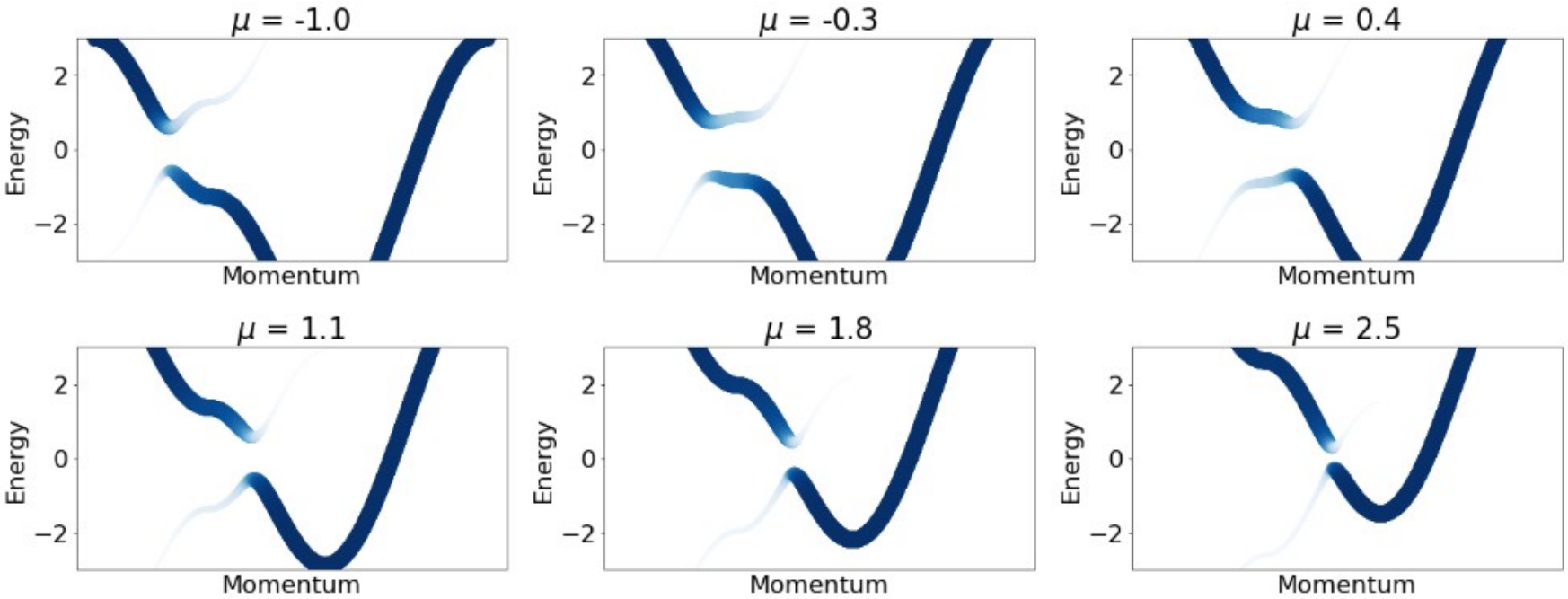
Impact of superconductivity in the electronic structure

A uniform superconducting terms opens up a gap in the electronic structure



Nodal superconductivity

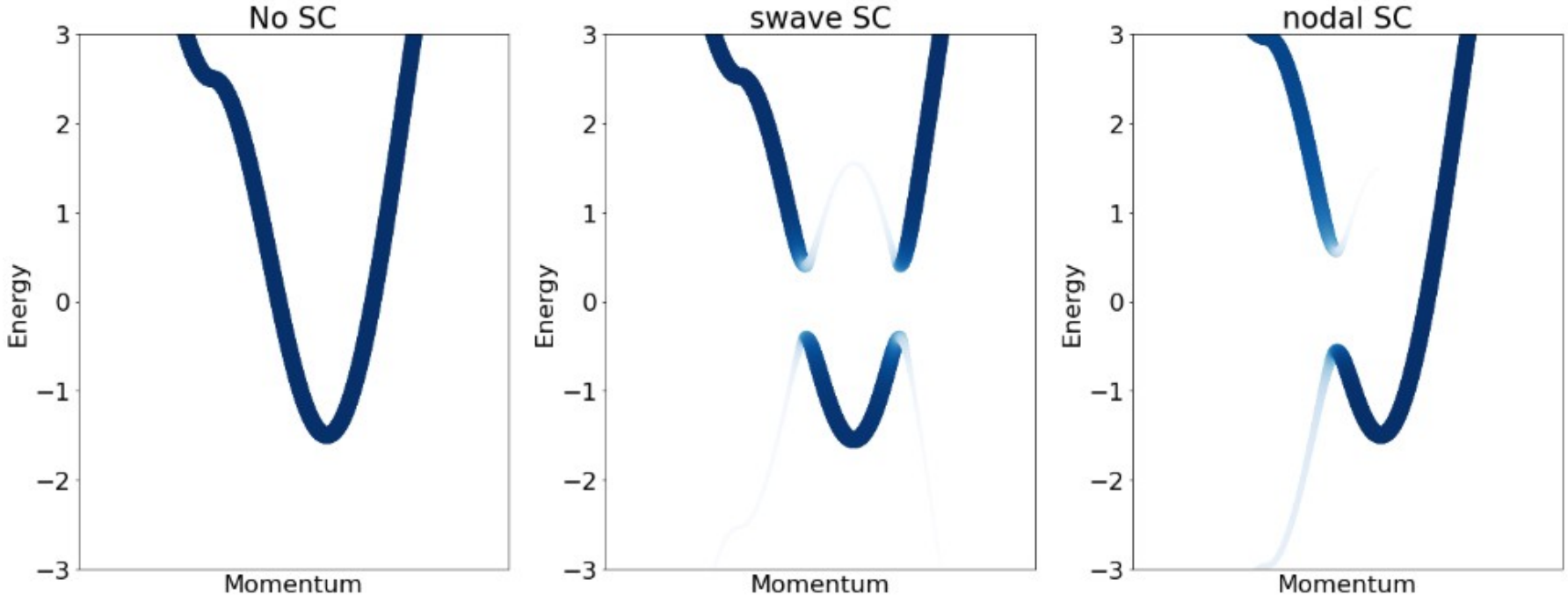
Let us now take a square lattice, and add a nodal superconducting order parameter



Some parts of the Fermi surface get a gap, while others remain gapless

Gapped and gapless superconductivity

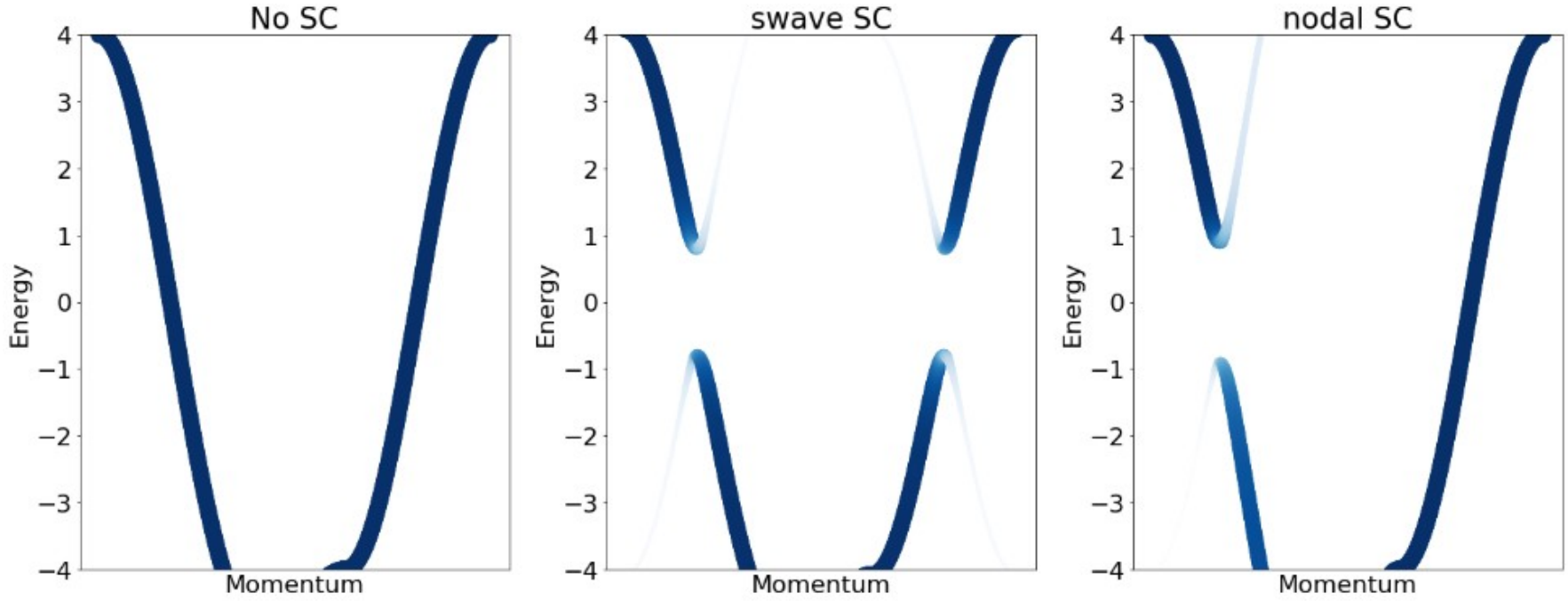
The electronic structure is modified differently depending on the type of superconductivity



Square lattice

Gapped and gapless superconductivity

The electronic structure is modified differently depending on the type of superconductivity



Triangular lattice

Unconventional superconductivity

Generic forms of superconductivity

A generic superconducting Hamiltonian

$$\widehat{H}' = \sum_{\boldsymbol{k},\sigma} \epsilon_{\boldsymbol{k}}\, \widehat{c}^{\dagger}_{\boldsymbol{k},\sigma}\, \widehat{c}_{\boldsymbol{k},\sigma} - \frac{1}{2}\sum_{\boldsymbol{k}}{}' \sum_{\sigma_1,\sigma_2} \left[\Delta_{\boldsymbol{k},\sigma_1\sigma_2}\, \widehat{c}^{\dagger}_{\boldsymbol{k},\sigma_1}\, \widehat{c}^{\dagger}_{-\boldsymbol{k},\sigma_2} + \Delta^{*}_{\boldsymbol{k},\sigma_1\sigma_2}\, \widehat{c}_{\boldsymbol{k},\sigma_1}\, \widehat{c}_{-\boldsymbol{k},\sigma_2}\right]$$

Can be characterized by a superconducting matrix

$$\Delta_{\boldsymbol{k}} = \begin{pmatrix} \Delta_{\boldsymbol{k},\uparrow\uparrow} & \Delta_{\boldsymbol{k},\uparrow\downarrow} \\ \Delta_{\boldsymbol{k},\downarrow\uparrow} & \Delta_{\boldsymbol{k},\downarrow\downarrow} \end{pmatrix}$$

The symmetry of the SC order determines the nature of the SC order

Superconducting momentum symmetries

A generic type of a superconductor is characterized by the order parameter

Real space

$$\Delta_{\uparrow\downarrow}(\mathbf{r},\mathbf{r}') \sim \langle c_{\mathbf{r}\uparrow} c_{\mathbf{r}'\downarrow} \rangle$$

Reciprocal space

$$\Delta_{\uparrow\downarrow}(\mathbf{k}) \sim \langle c_{\mathbf{k}\uparrow} c_{-\mathbf{k}\downarrow} \rangle$$

The superconducting state can be characterized by the symmetry of $\Delta_{\uparrow\downarrow}(\mathbf{k})$

Singlet and triplet superconductors

The superconducting order inherits a symmetry property

$$\Delta_{\vec{k},s_1 s_2} = -\Delta_{-\vec{k},s_2 s_1} = \begin{cases} \Delta_{-\vec{k},s_1 s_2} = -\Delta_{\vec{k},s_2 s_1} & \text{even} \\ -\Delta_{-\vec{k},s_1 s_2} = \Delta_{\vec{k},s_2 s_1} & \text{odd} \end{cases}$$

Spin-singlet (even)

$$\Delta_{\uparrow\downarrow}(\mathbf{k}) = \Delta_{\uparrow\downarrow}(-\mathbf{k})$$

Spin-triplet (odd)

$$\Delta_{\uparrow\uparrow}(\mathbf{k}) = -\Delta_{\uparrow\uparrow}(-\mathbf{k})$$

The symmetry of the superconducting order characterizes the superconductor

Generating a spin-triplet superconductor

Let us take a Hamiltonian breaking time-reversal with attractive interactions

$$H = t\sum_{\langle ij\rangle} c^{\dagger}_{i,s}c_{j,s} + J_z\sum_{i}\sigma_z^{s,s'}c^{\dagger}_{i,s}c_{j,s'} + V_1\sum_{\langle ij\rangle}\sum_{s}(c^{\dagger}_{i,s}c^{\dagger}_{i,s})\sum_{s'}(c^{\dagger}_{j,s'}c^{\dagger}_{j,s'})$$

kinetic *exchange* *attractive interactions*

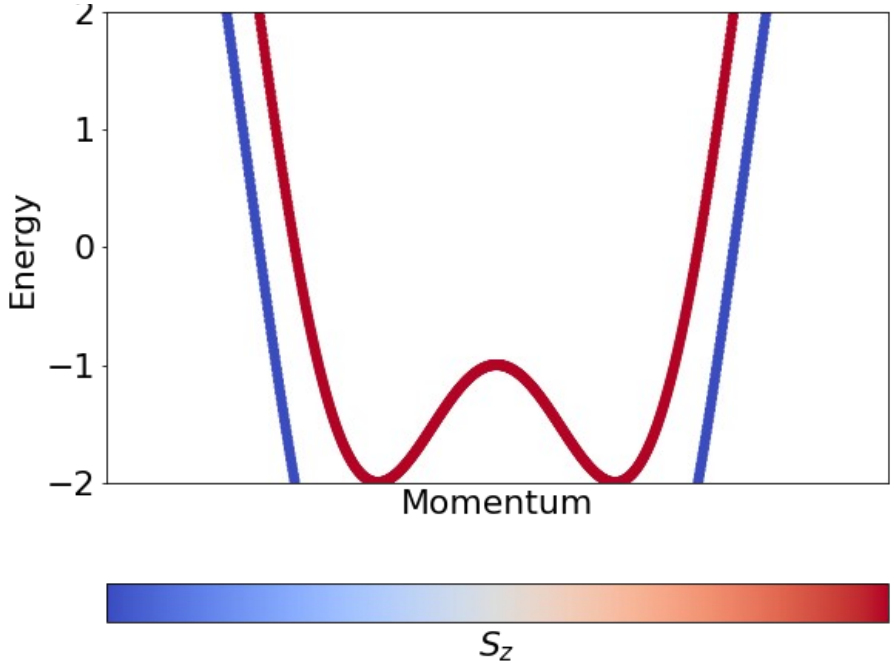
At the mean-field level, this may generate

$$H \sim \sum_{\langle ij\rangle}\Delta_{\uparrow\uparrow}c^{\dagger}_{i,\downarrow}c^{\dagger}_{j,\downarrow} + h.c. + ...$$

Generating a spin-triplet superconductor

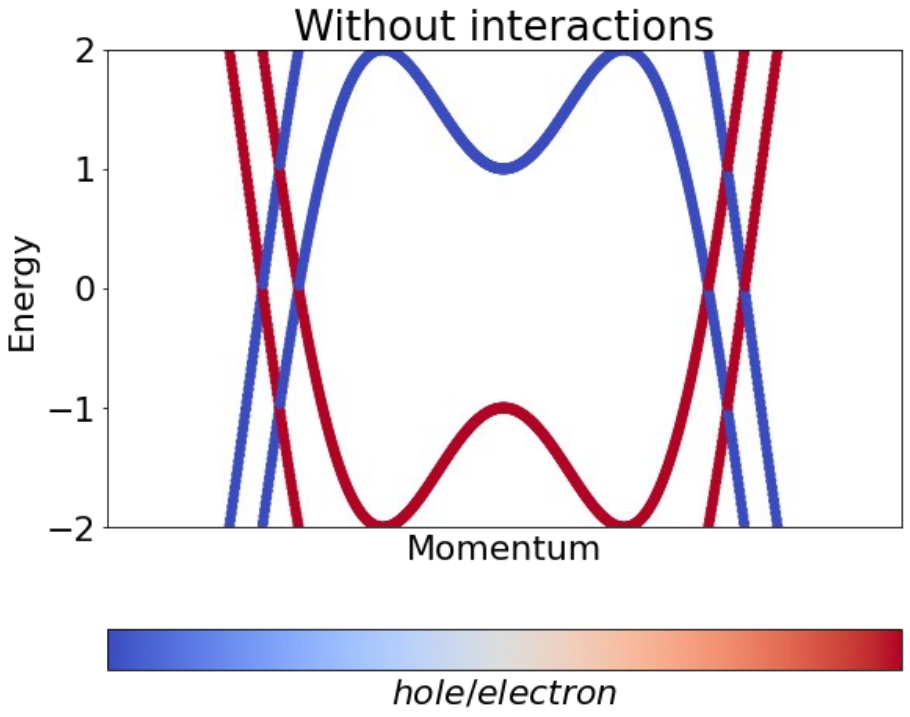
We start with a ferromagnetic 2D material, and see if interactions create superconductivity

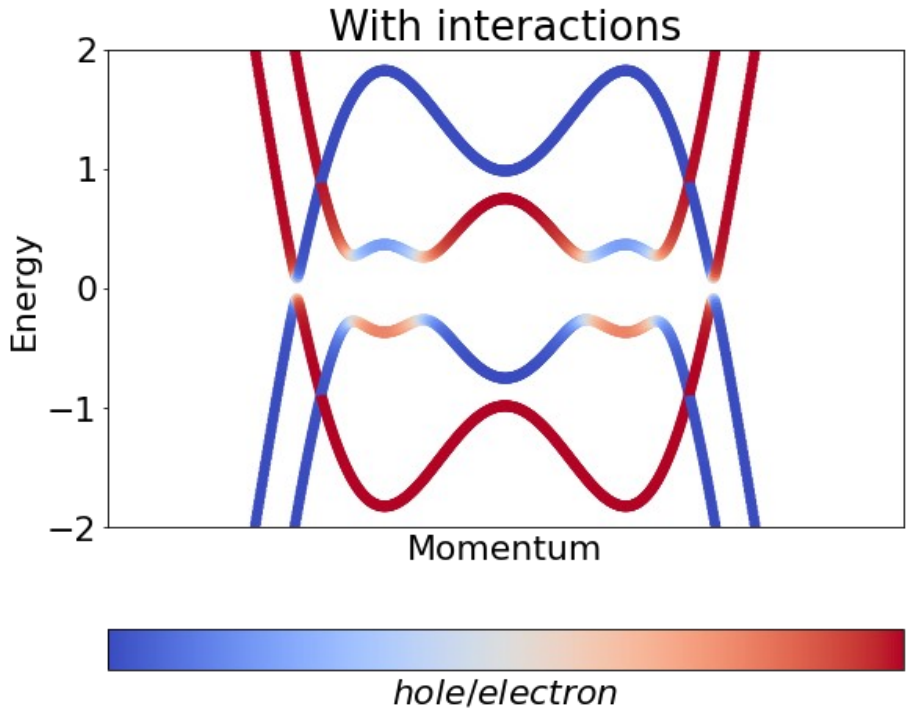
$$H = t \sum_{\langle ij \rangle} c^{\dagger}_{i,s} c_{j,s} + J_z \sum_{i} \sigma_z^{s,s'} c^{\dagger}_{i,s} c_{j,s'}$$



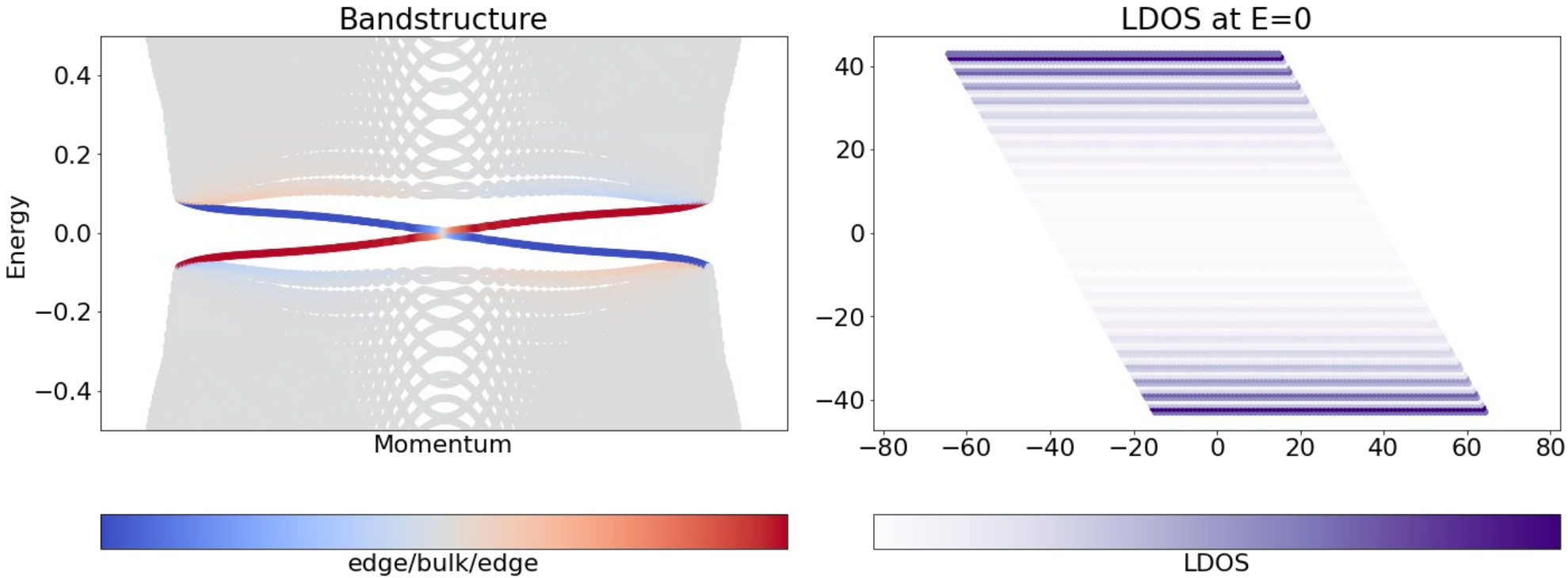
A spin-triplet superconductor

$$H = t\sum_{\langle ij\rangle} c^{\dagger}_{i,s}c_{j,s} + J_z\sum_{i}\sigma_z^{s,s'}c^{\dagger}_{i,s}c_{j,s'} + V_1\sum_{\langle ij\rangle}\sum_{s}(c^{\dagger}_{i,s}c^{\dagger}_{i,s})\sum_{s'}(c^{\dagger}_{j,s'}c^{\dagger}_{j,s'})$$





A spin-triplet superconductor, in a strip



A ferromagnetic superconductor can develop edge modes

Gapped and gapless superconductors

Let us focus on the superconducting order $\Delta_{\uparrow\downarrow}(\mathbf{k})$

Fully gapped

$$|\Delta_{\uparrow\downarrow}(\mathbf{k})|^2 > 0$$

Conventional superconductors (Pb, Al)

Gapless

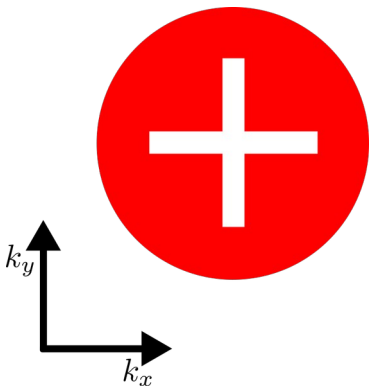
$$|\Delta_{\uparrow\downarrow}(\mathbf{k}_\alpha)|^2 = 0$$

Cuprate high-Tc superconductors

Superconducting momentum symmetries

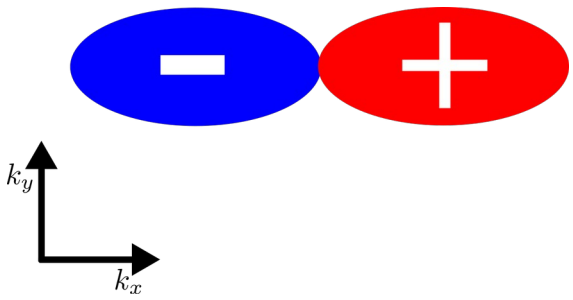
The superconducting state can be characterized by the symmetry of $\Delta(\mathbf{k})$

s-wave



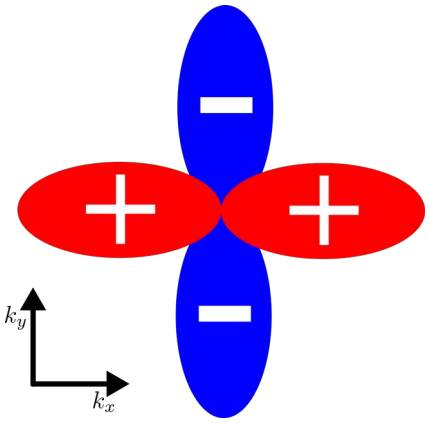
Conventional SC
(driven by phonons)

p-wave



Unconventional SC
(driven by FE magnons)

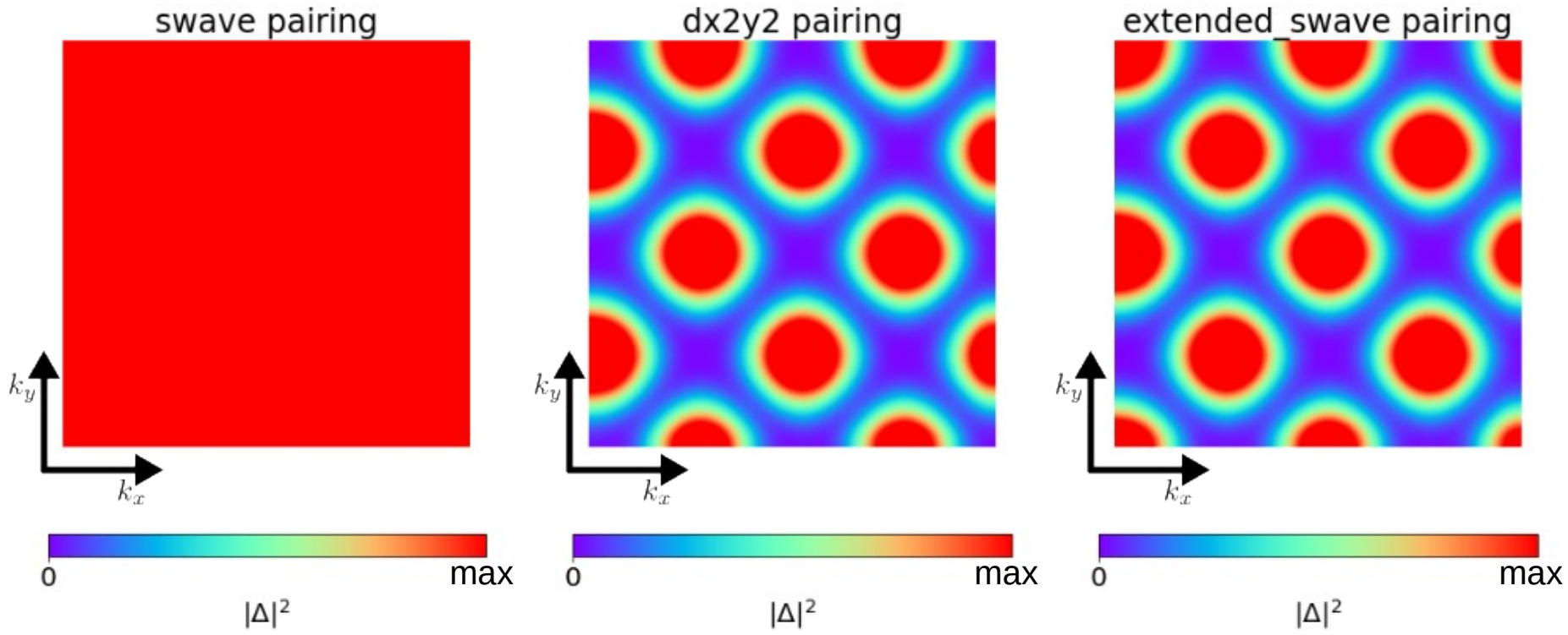
d-wave



Unconventional SC
(driven by AF magnons)

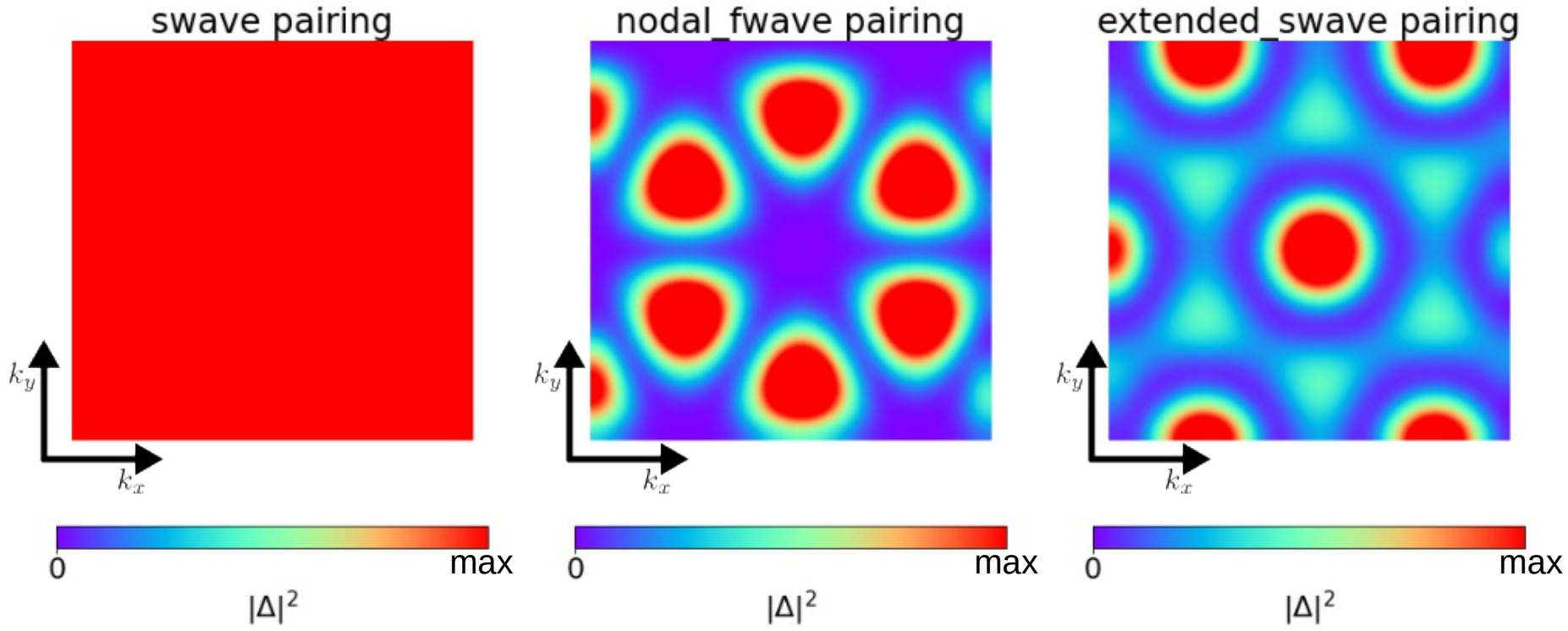
Some superconducting symmetries in the square lattice

Some superconducting symmetries in the square lattice

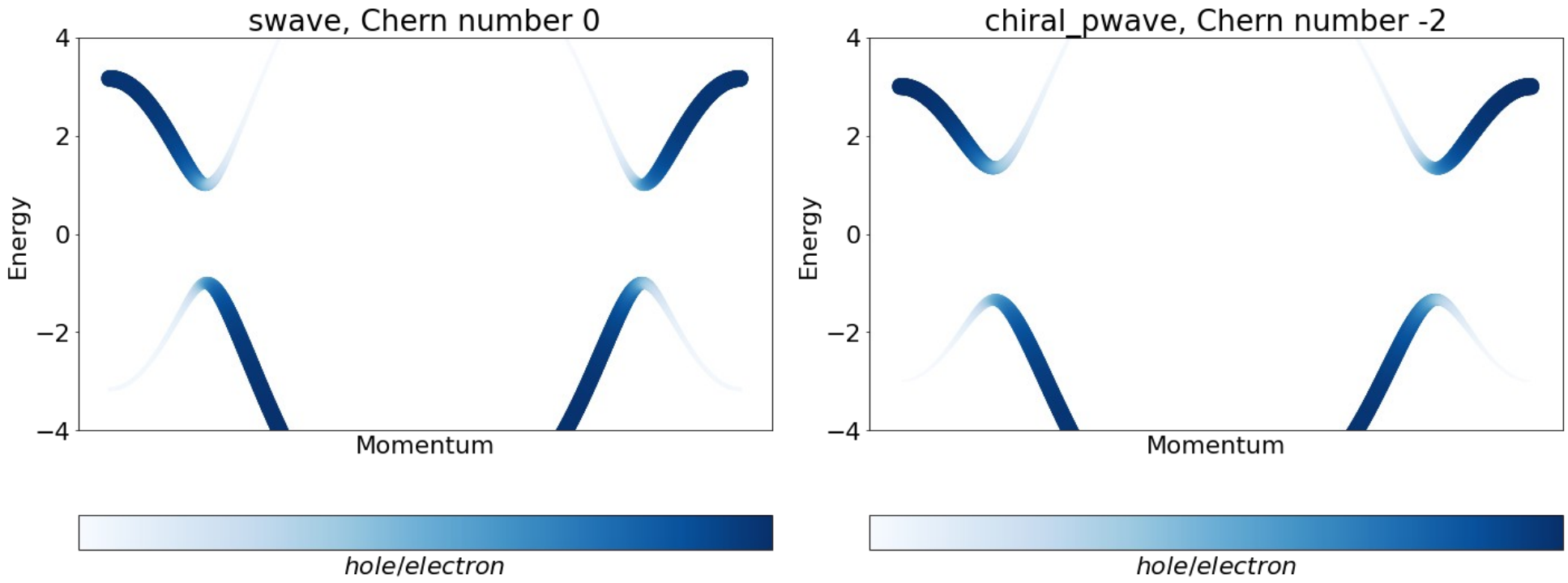


Some superconducting symmetries in the triangular lattice

Some superconducting symmetries in the triangular lattice

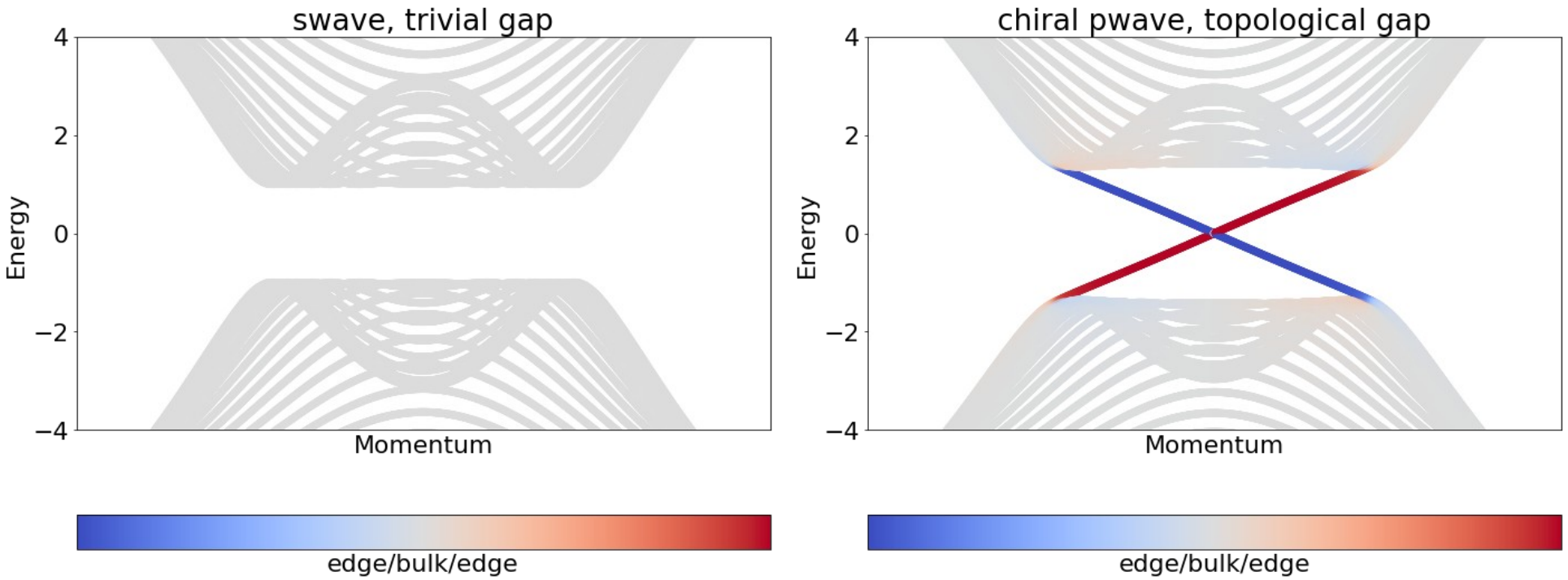


Gapped superconducting orders can be radically different



While both orders are gapped, they have different topological properties

Gapped superconducting orders can be radically different



The topological superconducting gap leads to protected edge excitations

Majorana physics in superconductors

The Nambu representation

How do we solve a Hamiltonian of the form

$$H = \sum_{\mathbf{k},s} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k},s} c_{\mathbf{k},s} + \sum_{\mathbf{k}} \Delta c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow}$$

Define a Nambu spinor

$$\Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{\mathbf{k}\downarrow} \\ c^{\dagger}_{-\mathbf{k}\downarrow} \\ -c^{\dagger}_{-\mathbf{k}\uparrow} \end{pmatrix}$$

Electron sector

Hole sector

The Hamiltonian in the Nambu basis is quadratic and can be diagonalized

$$H = \Psi^{\dagger}_{\mathbf{k}} \mathcal{H} \Psi_{\mathbf{k}}$$

Majorana excitations

A very special type of fermion is a so-called Majorana fermion

$$\Psi^\dagger = \Psi$$

Which by definition it is its own antiparticle

Majorana fermion do not appear naturally in materials, as we only have electrons

Yet, mathematically, each electron can be written as two Majoranas

$$c = \Psi_\alpha + i\Psi_\beta \qquad c^\dagger = \Psi_\alpha - i\Psi_\beta \qquad \Psi_\alpha^\dagger = \Psi_\alpha \qquad \Psi_\beta^\dagger = \Psi_\beta$$

Can we isolate a single Majorana in a material?

Looking at Majorana excitations in superconductors

Excitations in superconductors are combinations of electrons and holes, for instance

$$\Psi \sim c_n + c_n^\dagger$$

But this excitation is by definition a Majorana fermion

$$\Psi^\dagger = \Psi$$

Can we have superconductors in nature that show these excitations?

The minimal model for a 1D topological superconductor

One dimensional spinless p-wave superconductor (Kitaev model)

$$H = \sum_n t c_{n+1}^\dagger c_n + \Delta c_n c_{n+1} + c.c.$$

p-wave superconductivity

$$\Delta(k) = -\Delta(-k)$$

$$c_k = \sum_n e^{ikn} c_n$$

$$H = \sum_k \epsilon_k c_k^\dagger c_k + i\Delta \left[c_{-k} c_k \sin k - c_{-k}^\dagger c_k^\dagger \sin k \right]$$

A model Hamiltonian for topological superconductivity

Spinless fermions in a 1D chain (Kitaev model)

$$H = \sum_n c_n^\dagger c_{n+1} + c_n c_{n+1} + h.c.$$

Can be transformed into

$$c_n = \gamma_{2n-1} + i\gamma_{2n}$$

$$H = i\sum_n \gamma_{2n}\gamma_{2n+1}$$

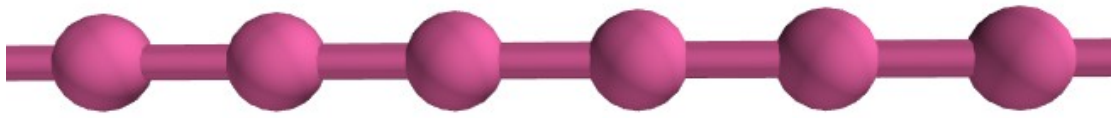
γ **Majorana operators**



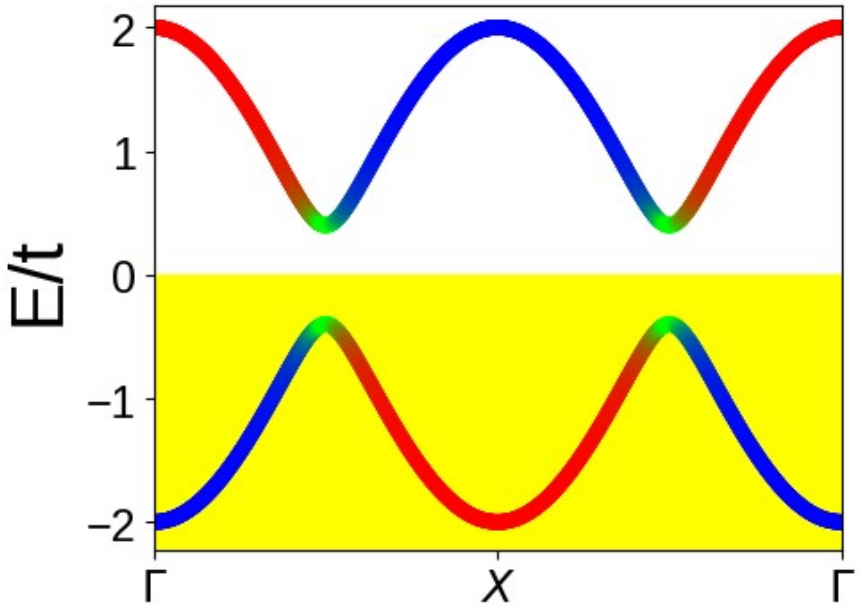
Unpaired Majorana operators

A model Hamiltonian for topological superconductivity

$$H = \sum_n c_n^\dagger c_{n+1} + \Delta c_n c_{n+1} + h.c.$$

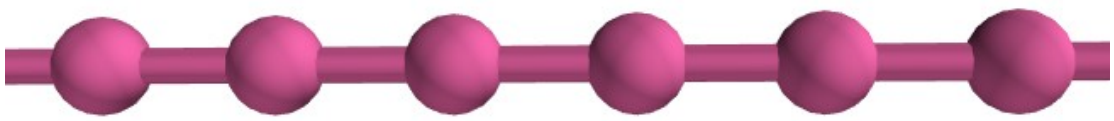


Infinite one dimensional chain

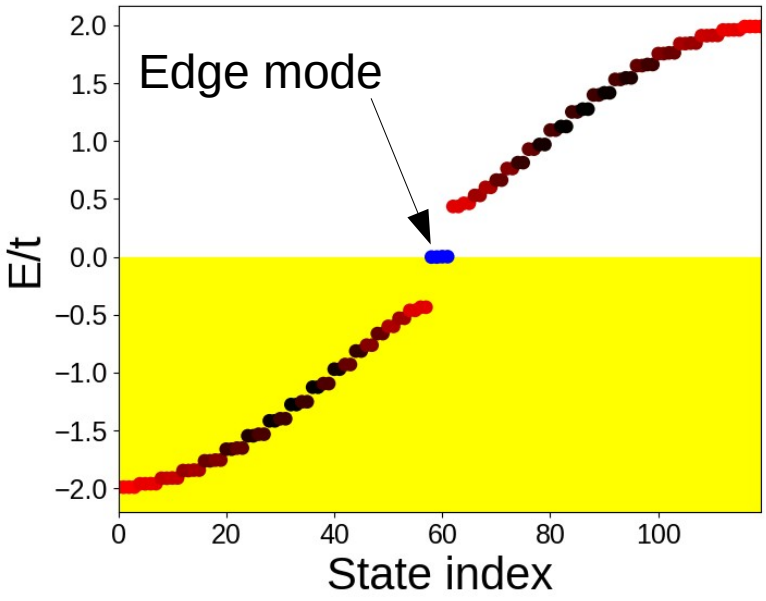


A model Hamiltonian for topological superconductivity

$$H = \sum_n c_n^\dagger c_{n+1} + \Delta c_n c_{n+1} + h.c.$$



Finite one dimensional chain



For finite systems, topological superconductivity gives rise to zero modes

Majorana states as topological surface modes

Generalized Kitaev model

$$H = \sum_n c_n^\dagger c_{n+1} + \Delta c_n c_{n+1} + \mu c_n^\dagger c_n + h.c.$$

p-wave SC

Chemical potential

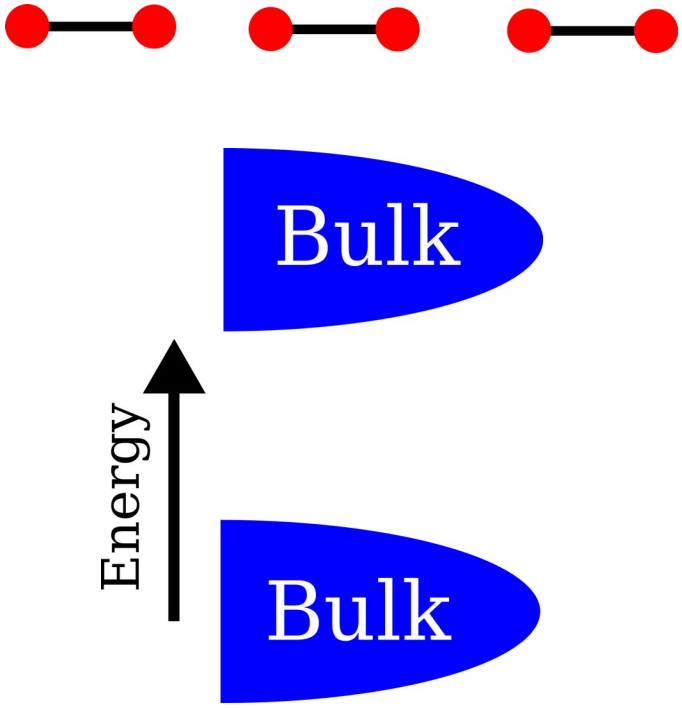
- Large chemical potential render the system filled, and topologically trivial
- p-wave SC promotes a topological phase with Majorana states

The emergence of Majorana states is associated to non-trivial topology

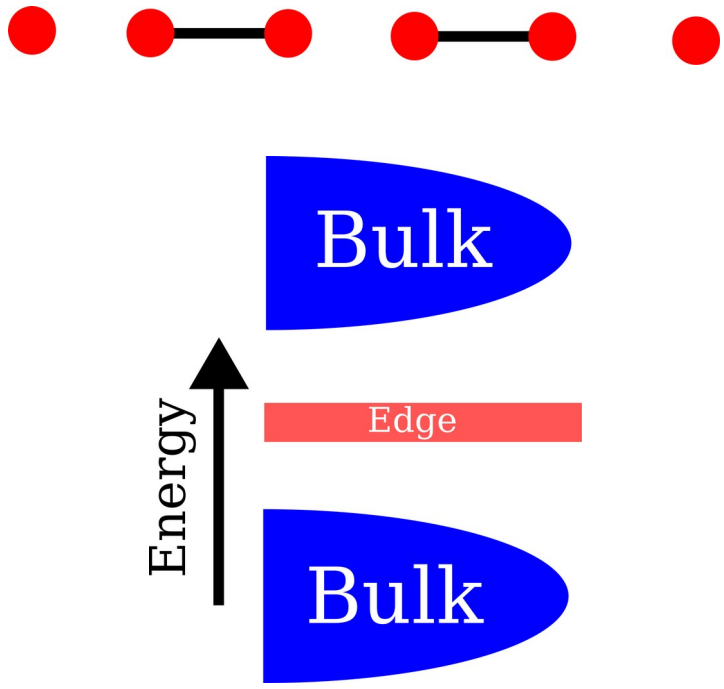
The two phases of the Kitaev model

(in the Majorana representation)

Trivial

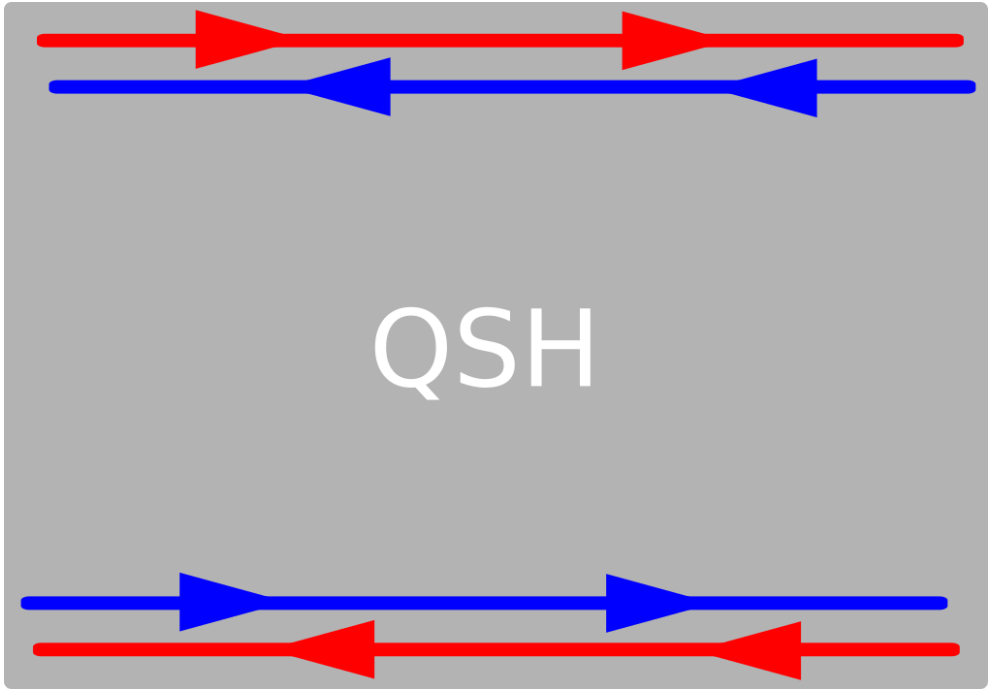


Topological



A simple way of building a topological superconductor with Majorana modes

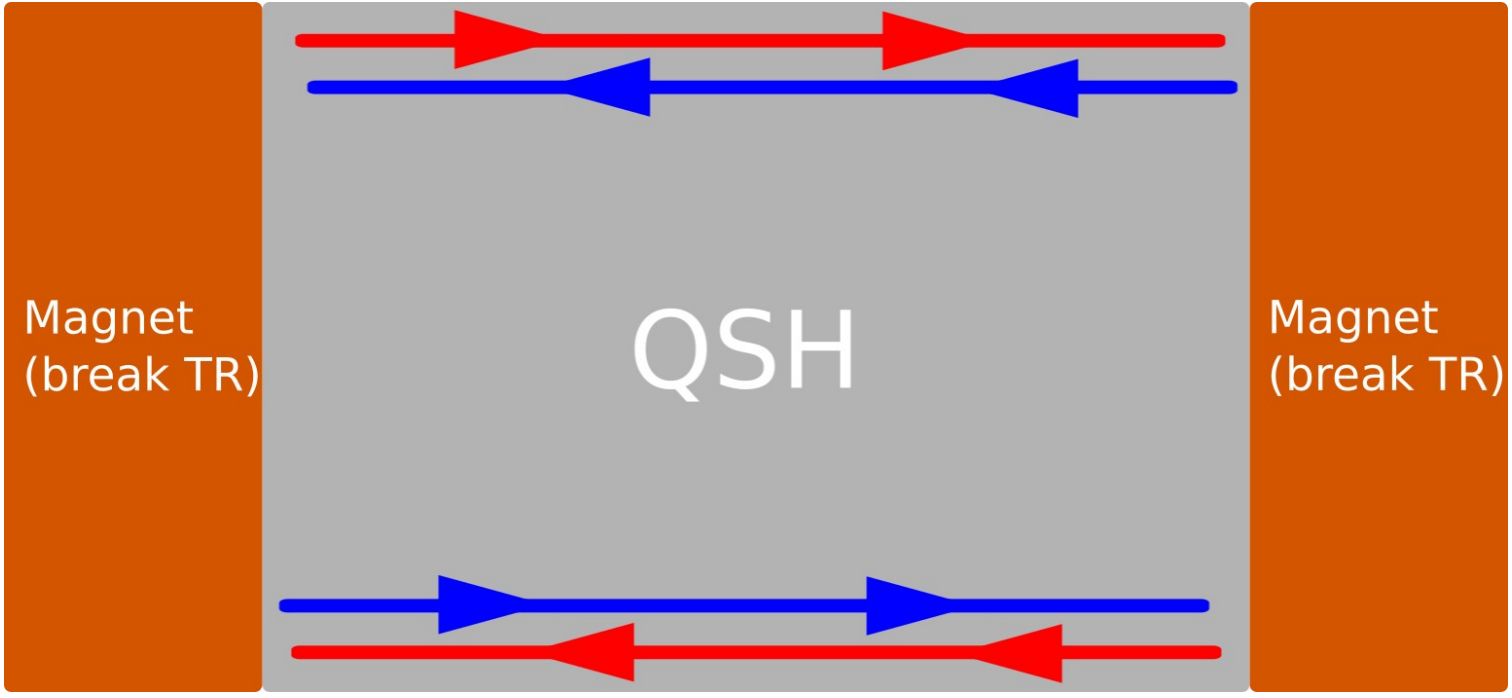
Taking the surface states of a quantum spin-Hall (QSH) insulator



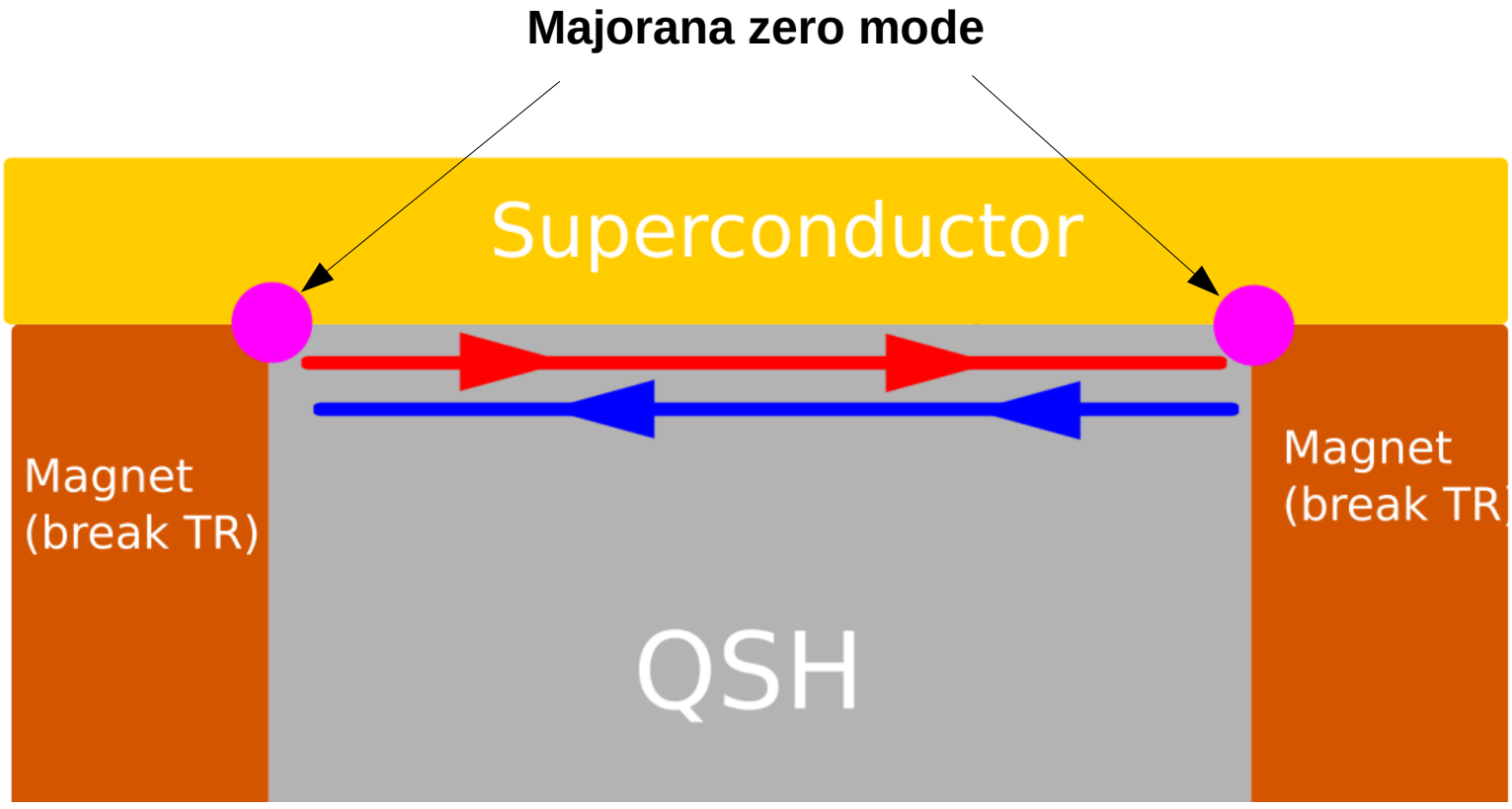
WTe_2

A simple way of building a topological superconductor with Majorana modes

Gap some of the helical modes with a magnet



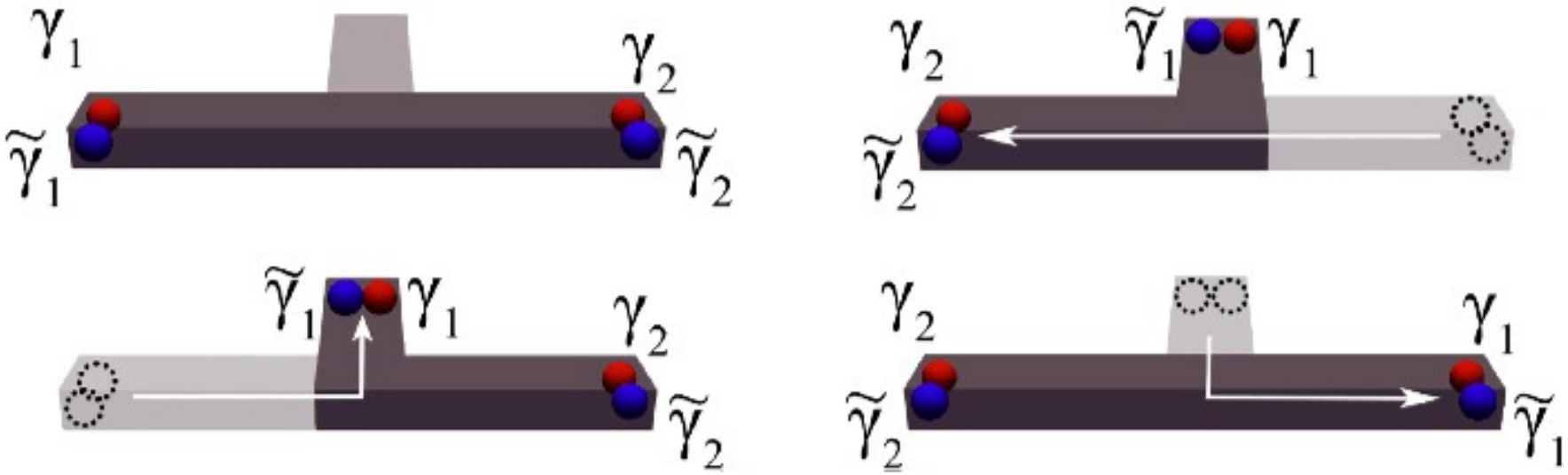
A simple way of building a topological superconductor with Majorana modes



Topological quantum computing

Topological quantum computing with Majoranas

Moving Majorana excitations around each other allows to perform quantum computations



Topological quantum computing with Majoranas

Imagine you have a system with 4 Majoranas

Then we can build 2 artificial fermionic operators as

$$c_\alpha \sim \gamma_1 + i\gamma_2 \qquad c_\beta \sim \gamma_3 + i\gamma_4$$

And our ground state manifold is 4-fold degenerate

$$|\Omega\rangle \qquad c_\alpha^\dagger|\Omega\rangle \qquad c_\beta^\dagger|\Omega\rangle \qquad c_\alpha^\dagger c_\beta^\dagger|\Omega\rangle$$

Can we use this degenerate manifold as the space for a quantum computation?

Topological quantum computing with Majoranas

Let us take that now we wind 2 out those 4 Majoranas

γ_1, γ_2 (for example)

The ground state gets transformed as

$$U_{12} = e^{i\frac{\pi}{4}\gamma_1\gamma_2}$$

Most importantly, the switching operators define a non-abelian algebra

$$U_{12}U_{34} \neq U_{34}U_{12}$$

Switching Majoranas allows representing different quantum algorithms using topologically protected degenerate many-body states

Take home

- Superconductivity arises from attractive interactions
- Superconducting states can be captured by effective single-particle mean-field models
- Unconventional superconductors can show Majorana excitations